

Entrega TII

Metódos Matemáticos Para a Física

1. Dada la semicircunferencia dada por la parametrización

$$c: [0, \pi] \rightarrow \mathbb{R}^3, \theta \mapsto (0, a \sin \theta, a \cos \theta); a > 0$$

Supóngase que dicha semicircunferencia esta hecha de un alambre con una densidad uniforme de 2 gramos por unidad de longitud.

- a) ¿Cuál es la masa total del alambre?
- b) ¿Dónde está el centro de masa del alambre?

$$\begin{aligned} x(\theta) &= 0 & \theta \in [0, \pi] \\ y(\theta) &= a \sin \theta \\ z(\theta) &= a \cos \theta \end{aligned}$$

$$a) \rho = 2 \text{ g/m} \quad L = \pi a \rightarrow M = 2\pi a$$

$$b) ds = \sqrt{x'(\theta)^2 + y'(\theta)^2 + z'(\theta)^2} d\theta \rightarrow = \sqrt{0^2 + a^2 \cos^2 \theta + a^2 \sin^2 \theta} : a$$

$$z_{cm} = \frac{1}{M} \int_C \rho \cdot z \, ds = \frac{1}{2\pi a} \int_0^\pi 2a \cos \theta \cdot a \, d\theta = \frac{1}{2\pi a} \int_0^\pi 2a^2 \cos \theta \, d\theta = \frac{1}{2\pi a} [2a^2 \sin(\theta)]_0^\pi$$

$$z_{cm} = 0$$

Como la semicircunferencia esta definida sobre yz $\rightarrow s_{cm} = (0, 0, 0)$

2. Hallar la masa de un alambre que sigue la intersección de la esfera $x^2 + y^2 + z^2 = 1$ y el plano $x + y + z = 0$ si la densidad (x, y, z) esta dada por $\rho(x, y, z) = x^2$ gramos por unidad de longitud del alambre.

$$M = \int \rho(x, y, z) ds$$

$$C: \begin{cases} x(t) = \frac{\sqrt{2}}{2} \cos t + \frac{\sqrt{6}}{6} \sin t \\ y(t) = -\frac{\sqrt{2}}{2} \cos t + \frac{\sqrt{6}}{6} \sin t \\ z(t) = -\frac{\sqrt{6}}{3} \sin t \end{cases} \quad t \in [0, 2\pi]$$

$$C'(t) = \begin{cases} x'(t) = -\frac{\sqrt{2}}{2} \sin t + \frac{\sqrt{6}}{6} \cos t \\ y'(t) = \frac{\sqrt{2}}{2} \sin t + \frac{\sqrt{6}}{6} \cos t \\ z'(t) = -\frac{\sqrt{6}}{3} \cos t \end{cases}$$

$$ds = \sqrt{x'^2(t) + y'^2(t) + z'^2(t)} dt = \sqrt{\left(\frac{1}{2} \sin^2 t + \frac{1}{6} \cos^2 t\right) + \left(\frac{1}{2} \sin^2 t + \frac{1}{6} \cos^2 t\right) + \frac{2}{3} \cos^2 t}$$

$$ds = \sqrt{\sin^2 t + \cos^2 t} = 1$$

$$\int_C x^2 ds = \int_0^{2\pi} \left(\frac{\sqrt{2}}{2} \cos t + \frac{\sqrt{6}}{6} \sin t \right)^2 dt = \int_0^{2\pi} \frac{1}{2} \cos^2 t + \frac{1}{6} \sin^2 t + \frac{\sqrt{3}}{6} \cos t \sin t$$

$$= \int_0^{2\pi} \frac{\cos 2t + 1}{4} + \frac{1 - \cos 2t}{12} + \frac{\sqrt{3}}{12} \sin 2t dt = \int_0^{2\pi} \frac{3 \cos 2t + 3}{12} + \frac{1 - \cos 2t}{12} + \frac{\sqrt{3}}{12} \sin 2t$$

$$= \frac{1}{12} \int_0^{2\pi} 2 \cos 2t + \sqrt{3} \sin 2t + 4 dt = \frac{1}{12} \left[\sin 2t - \frac{\sqrt{3}}{2} \cos 2t + 4t \right]_0^{2\pi} = \frac{8\pi}{12} = \frac{2\pi}{3}$$

3. Sea $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, calcular la integral de $\mathbf{F}$ a lo largo de la trayectoria $\mathbf{c}(t) = (t, t, t)$, $0 \leq t \leq 1$

$$\mathbf{r}(t) = t\vec{i} + t\vec{j} + t\vec{k} \quad \int_C \mathbf{F}(x, y, z) ds = \int_0^1 \mathbf{F}(x(t), y(t), z(t)) \cdot \mathbf{r}'(t) dt$$

$$\mathbf{r}'(t) = \vec{i} + \vec{j} + \vec{k}$$

$$\int_C \mathbf{F}(x, y, z) ds = \int_0^1 (t, t, t) \cdot (1, 1, 1) dt = \int_0^1 3t dt = \left[\frac{3t^2}{2} \right]_0^1 = \frac{3}{2}$$

4. Evaluar la integral de línea $\int_C x^2 dx - xy dy + dz$ donde C es la parábola $z = x^2, y = 0$ desde $(-1, 0, 1)$ hasta $(1, 0, 1)$.

$$\mathbf{r}(t) = \begin{cases} x(t) = t \\ y(t) = 0 \\ z(t) = t^2 \end{cases} \quad t \in [-1, 1] \quad \mathbf{f}(x, y, z) = (x^2, -xy, 1)$$

$$\int_C \mathbf{F}(x, y, z) ds = \int_{-1}^1 \mathbf{F}(x(t), y(t), z(t)) \cdot \mathbf{r}'(t) dt = \int_{-1}^1 (t^2, 0, 1) \cdot (1, 0, 2t) dt$$

$$= \int_{-1}^1 t^2 + 2t dt = \left[\frac{t^3}{3} + t^2 \right]_{-1}^1 = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}$$

5. Encontrar el plano tangente a la superficie

a) $x = u^2, y = u \operatorname{sen} e^v, z = \frac{1}{3}u \cos e^v$, en $(13, -2, 1)$

b) $z = 3x^2 + 8xy, x = 1, y = 0$

a) $r(u, v) = (u^2, u \operatorname{sen} e^v, \frac{1}{3}u \cos e^v)$

$$r_u = (2u, \operatorname{sen} e^v, \frac{1}{3} \cos e^v)$$

$$r_v = (0, u e^v \cos(e^v), -\frac{1}{3} u e^v \operatorname{sen}(e^v))$$

Vector normal del plano: $\vec{n} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2u & \operatorname{sen} e^v & \frac{1}{3} \cos e^v \\ 0 & u e^v \cos e^v & -\frac{1}{3} u e^v \operatorname{sen} e^v \end{vmatrix}$

$$\vec{n} = -\frac{1}{3} u e^v \operatorname{sen}^2 e^v \vec{i} + 2u^2 e^v \cos e^v \vec{j} - \frac{1}{3} u e^v \cos^2 e^v \vec{k} + \frac{2}{3} u^2 e^v \operatorname{sen} e^v \vec{j}$$

$$\vec{n} = -\frac{1}{3} u e^v \vec{i} + \frac{2}{3} u^2 e^v \operatorname{sen} e^v \vec{j} + 2u^2 e^v \cos e^v \vec{k} = -\frac{1}{3} u e^v (1, 2 \operatorname{sen} e^v, -6 u \operatorname{sen} e^v)$$

$$x = u^2 \rightarrow u^2 = 13 \quad u = \sqrt{13}$$

$$y = \sqrt{13} \operatorname{sen} e^v \rightarrow \operatorname{sen} e^v = \frac{-2}{\sqrt{13}}$$

$$z = \sqrt{13}/3 \cos e^v \rightarrow \cos e^v = \frac{3}{\sqrt{13}}$$

e^v no afecta el vector normal

$$\vec{n} = -\frac{\sqrt{13}}{3} (1, -2 \cdot \sqrt{13} \cdot \frac{2}{\sqrt{13}}, -6 \cdot \sqrt{13} \cdot \frac{3}{\sqrt{13}}) \Rightarrow \vec{n} = (1, 4, -18)$$

$$\Pi = Ax + By + Cz + D = 0 \quad x + 4y - 18z + D = 0 \rightarrow (x-13) + 4(y+4) - 18(z-1) = 0$$

$$x - 13 + 4y + 16 - 18z + 18 = 0 \rightarrow x + 4y - 18z + 13 = 0 \rightarrow \text{Plano Tgte.}$$

$$b) \quad z = 3x^2 + 8xy \quad x = 1, y = 0$$

$$z = 3 \cdot 1^2 = 3$$

$$f(x, y) = z \rightarrow \begin{aligned} f_x &= 6x + 8y = (6, 8) \rightarrow E_u(1, 0) \rightarrow f_x = 6 \\ f_y &= 8x = (8, 0) \rightarrow E_u(1, 0) \rightarrow f_y = 8 \end{aligned}$$

$$\vec{n} = (-f_x, -f_y, 1) = (6, 8, -1)$$

$$6x + 8y - z + 1 = 0 \rightarrow 6(x-1) + 8(y-0) - (z-3) = 0 \rightarrow 6x + 8y - z - 3 = 0$$

6. Encontrar la expresión de un vector unitario normal a la superficie

$$x = \sin v, y = u, z = \cos v \quad \text{para} \quad 0 \leq v \leq 2\pi \text{ y } -1 \leq u \leq 3$$

$$r(u, v) = (\sin v, u, \cos v)$$

$$r_u = (0, 1, 0)$$

$$r_v = (\cos v, 0, -\sin v)$$

$$\vec{n} = r_u \times r_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 0 \\ \cos v & 0 & -\sin v \end{vmatrix} = -\sin(v) \vec{i} - \cos v \vec{k}$$

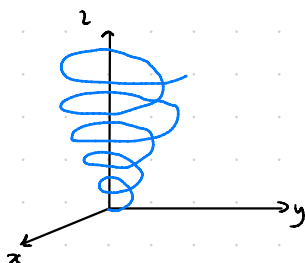
$$\vec{n} = (-\sin v, 0, -\cos v)$$

7. Considerar la superficie en R^3 parametrizada por

$$\Phi(r, \theta) = (r \cos \theta, r \sin \theta, \theta), \quad 0 \leq r \leq 1 \quad 0 \leq \theta \leq 4\pi$$

- Dibujar y describir la superficie.
- Hallar una expresión para una normal unitaria a la superficie.
- Hallar una ecuación para el plano tangente a la superficie en el punto (x_0, y_0, z_0) .
- Si (x_0, y_0, z_0) es un punto de la superficie, demostrar que el segmento de línea horizon-

a)



Helice que va aumentando radialmente de 0 a 1

b)

$$\begin{aligned}\Phi_r &= (\cos \theta, \sin \theta, 0) \\ \Phi_\theta &= (-r \sin \theta, r \cos \theta, 1)\end{aligned}$$

$$\vec{n} = \Phi_r \times \Phi_\theta = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos \theta & \sin \theta & 0 \\ -r \sin \theta & r \cos \theta & 1 \end{vmatrix} = \sin \theta \vec{i} + r \cos^2 \theta \vec{j} + r \sin^2 \theta \vec{k}$$

$$\vec{n} = (\sin \theta, -\cos \theta, r)$$

$$\hat{n} = \frac{\vec{n}}{\|\vec{n}\|} \quad \|\vec{n}\| = \sqrt{\sin^2 \theta + \cos^2 \theta + r^2} = \sqrt{1 + r^2}$$

$$\hat{n} = \frac{\sin \theta \vec{i} - \cos \theta \vec{j} + r \vec{k}}{\sqrt{1 + r^2}}$$

$$\begin{aligned}x_0 &= r_0 \cos \theta_0 \\ y_0 &= r_0 \sin \theta_0 \\ z_0 &= \theta_0\end{aligned}$$

$$c) n_{\theta} = (\sin \theta, -\cos \theta, r) \rightarrow$$

$$\nabla \equiv \sin \theta_0 x - \cos \theta_0 y + r_0 z + D = 0$$

$$\sin \theta_0 (x - r_0 \cos \theta_0) - \cos \theta_0 (y - r_0 \sin \theta_0) + r_0 (z - \theta_0) = 0$$

$$\sin \theta_0 x - r_0 \cos \theta_0 \sin \theta_0 - \cos \theta_0 y + r_0 \cos \theta_0 \sin \theta_0 + r_0 z - r_0 \theta_0 = 0$$

$$\nabla \equiv \sin \theta_0 x - \cos \theta_0 y + r_0 z = r_0 \theta_0$$

8) Enunciado incompleto

8. Evaluar $\iint_S z \, dS$ donde S es la superficie $z = x^2 + y^2, x^2 + y^2 \leq 1$

$$f(x, y, z) = x^2 + y^2 = g(x, y)$$

$$\begin{aligned} g_x &= 2x \\ g_y &= 2y \end{aligned}$$

$$dS = \sqrt{1 + g_x^2 + g_y^2} \, dA$$

$$\iint_S f(x, y, z) \, dS = \iint f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dA$$

$$= \iint x^2 + y^2 \sqrt{1 + 4x^2 + 4y^2} \, dx \, dy$$

Polar coordinates: $\rightarrow \int_0^{2\pi} \int_0^1 r^2 \sqrt{1 + 4r^2} \cdot r \, dr \, d\theta = \int_0^{2\pi} \int_0^1 r^3 \sqrt{1 + 4r^2} \, dr \, d\theta$

$$\int_0^1 r^3 \sqrt{1 + 4r^2} \, dr = \int_0^1 \frac{t-1}{4} \cdot \sqrt{t} \cdot \frac{1}{8} dt = \frac{1}{32} \int_0^1 (t-1) \sqrt{t} \, dt = \int_0^1 t^{3/2} - t^{1/2} \, dt = \left[\frac{1}{2} t^{5/2} - \frac{2}{3} t^{3/2} \right]_0^1$$

$$\frac{t}{dt} = 1 + 4r^2 \rightarrow r^2 = \frac{t-1}{4}$$

$$= \frac{1}{32} \left[\frac{2}{5} (1+4r^2)^{5/2} - \frac{2}{3} (1+4r^2)^{3/2} \right]_0^1 = \frac{1}{32} \left[\left(\frac{2}{5} (5)^{5/2} - \frac{2}{3} (5)^{3/2} \right) - \left(\frac{2}{5} - \frac{2}{3} \right) \right]$$

$$= \frac{1}{32} \left[\frac{2}{5} (5)^{5/2} - \frac{2}{3} (5)^{3/2} + \frac{4}{15} \right] = \frac{1}{80} (5)^{5/2} - \frac{1}{48} (5)^{3/2} + \frac{1}{120}$$

$$\int_0^{2\pi} \left(\frac{25\sqrt{5}}{80} - \frac{5\sqrt{5}}{48} + \frac{1}{120} \right) d\theta = \int_0^{2\pi} \left(\frac{5\sqrt{5}}{24} + \frac{1}{120} \right) d\theta = \frac{1}{120} \int_0^{2\pi} (25\sqrt{5} + 1) d\theta = \frac{1}{60} (25\sqrt{5} + 1)$$

9. Hallar el área del disco D de radio R usando el teorema de Green.

$$\text{Teorema de Green} \rightarrow \int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\text{Aplicamos la idea de que } \iint_R M dx + N dy = \iint_R 1 dx dy \quad \text{si } \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 1$$

$$\iint_D 1 dx dy \rightarrow \text{Al ser un disco, pasamos a polares}$$

$$\begin{aligned} x &= r \cos \theta & r &\in [0, R] \\ y &= r \sin \theta & \theta &\in [0, 2\pi] \\ dx dy &= r dr d\theta \end{aligned}$$

$$\int_0^{2\pi} \int_0^R r dr d\theta = \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^R d\theta = \int_0^{2\pi} \frac{R^2}{2} d\theta = \left[\theta \frac{R^2}{2} \right]_0^{2\pi} = \pi R^2$$

10. Verificar el teorema de Green para el disco D con centro (0, 0) y radio R y la función

$$P(x, y) = x + y, Q(x, y) = y$$

$$\oint_C P dx + Q dy = \iint_R \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} dx dy$$

$$\frac{\partial Q}{\partial x} = 0 \quad \frac{\partial P}{\partial y} = 1 \quad \iint_R (0 - 1) dx dy = \iint_R -1 dx dy$$

$$\text{Polares} \rightarrow - \int_0^{2\pi} \int_0^R r dr d\theta = -\pi R^2$$

$$\oint x + y \, dx + y \, dy = \int_0^{2\pi} (R \cos \theta + R \sin \theta)(-R \sin \theta) + (R \cos \theta)(R \sin \theta) \, d\theta$$

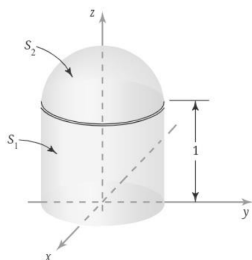
Polar coordinates: $x = R \cos \theta$ $dx = -R \sin \theta \, d\theta$
 $y = R \sin \theta$ $dy = R \cos \theta \, d\theta$

$$\int_0^{2\pi} -R^2 \cos \theta \sin \theta + R^2 \cos \theta \sin \theta - R^2 \sin^2 \theta \, d\theta = - \int_0^{2\pi} R^2 \sin^2 \theta \, d\theta = -\frac{R^2}{2} \int_0^{2\pi} 1 - \cos 2\theta = \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{2\pi}$$

$$= -\frac{R^2}{2} \cdot 2\pi = -\pi R^2$$

$$\oint_C P \, dx + Q \, dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA \quad \rightarrow \text{Stokes' theorem}$$

11. Sea S la superficie cilíndrica con tapa mostrada en la siguiente figura



S es la unión de dos superficies S_1 y S_2 , donde S_1 es el conjunto de (x, y, z) tales que

$$x^2 + y^2 = 1, 0 \leq z \leq 1$$

y S_2 es el conjunto de puntos (x, y, z) tales que

$$x^2 + y^2 + (z - 1)^2 = 1, z \geq 1$$

Sea el campo vectorial

$$\mathbf{F}(x, y, z) = (zx + z^2y + x)\mathbf{i} + (z^3yx + y)\mathbf{j} + z^4x^2\mathbf{k}.$$

Calcular $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ usando el teorema de Stokes.

$$\begin{aligned} 1) \nabla \times \mathbf{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f(x) & f(y) & f(z) \end{vmatrix} = 0\vec{i} + (x + 2zy)\vec{j} + z^3y\vec{k} - z^2\vec{h} - 3z^2yx\vec{c} - 2xz^4\vec{j} \\ &= -3z^2yx\vec{c} + (x + 2zy - 2xz^4)\vec{j} + (z^3y - z^2)\vec{h} \end{aligned}$$

Pero nos damos que es una superficie cerrada por ello.

$$\iint \nabla \mathbf{F} \cdot d\mathbf{S} = 0$$

12. Calcular $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ donde S es la semiesfera

$$x^2 + y^2 + z^2 = 1, x \geq 0$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{N} dS$$

$$\vec{F} = x^3 \mathbf{i} - y^3 \mathbf{j}$$

$$\vec{N} = (1, 1, 1) \quad \vec{\nabla} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^3 & -y^3 & 0 \end{vmatrix} = 0 + 0 + 0 - 0 - 0 - 0 = 0$$

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{N} dS = 0$$

13. Verificar el teorema de Stokes para el helicoido

$$\Phi(r, \theta) = (r \cos \theta, r \sin \theta, \theta), (r, \theta) \in [0, 1] \times [0, \pi/2]$$

y el campo vectorial $\mathbf{F}(x, y, z) = (z, x, y)$

$$\int_C \vec{F} \cdot d\mathbf{r} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{N} dS$$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \vec{i} + \vec{j} + \vec{k}$$

Calcular vector $\vec{N}$:

$$\Phi_r = (\cos \theta, \sin \theta, 0) \quad \Phi_\theta = (-r \sin \theta, r \cos \theta, 1) \quad \vec{N} = \Phi_r \times \Phi_\theta$$

$$\vec{N} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos \theta & \sin \theta & 0 \\ -r \sin \theta & r \cos \theta & 1 \end{vmatrix} = \sin \theta \vec{i} + r \cos^2 \theta \vec{k} + r \sin^2 \theta \vec{k} - \cos \theta \vec{j} \quad \vec{N} = (\sin \theta, -\cos \theta, r)$$

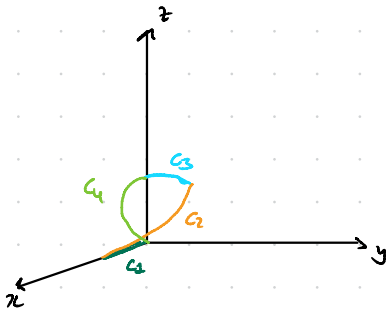
$$\iint_S (1, 1, 1) \cdot (\sin \theta, -\cos \theta, r) dS = \iint_S \sin \theta - \cos \theta + r dS$$

$$= \int_0^1 \int_0^{\pi/2} \sin \theta - \cos \theta + r dr d\theta \rightarrow \int_0^1 [-\cos \theta - \sin \theta + r\theta]_0^{\pi/2} dr = \int_0^1 -1 + \frac{\pi r}{2} + 1 dr = \int_0^1 \frac{\pi r}{2} dr$$

$$= \left[\frac{\pi r^2}{4} \right]_0^1 = \pi/4$$

Integral de línea

$$\int_C F \cdot dr = \int z dx + x dy + y dz$$



$$\int_C F \cdot dr = \int_{C1} + \int_{C2} + \int_{C3} + \int_{C4}$$

$$C1: \begin{cases} r_1(r) = (r, 0, 0) \\ C2: \begin{cases} r_2(\theta) = (\cos(\theta), \sin(\theta), 0) \\ C3: \begin{cases} r_3(r) = (0, r, \pi/2) \\ C4: \begin{cases} r_4(\theta) = (0, 0, \theta) \end{cases} \end{cases} \end{cases}$$

$$\int_{C1} F(x(t), y(t), z(t)) \cdot dr_1 = \int_0^1 (0, 0, 0) \cdot (1, 0, 0) dr = 0$$

$$\int_{C2} F(x(t), y(t), z(t)) \cdot dr_2 = \int_0^{\pi/2} (0, \cos \theta, \sin \theta) \cdot (-\sin \theta, \cos \theta, 1) d\theta = \int_0^{\pi/2} -\theta \sin \theta + \cos^2 \theta + \sin \theta d\theta$$

$$= \left[\theta \cos \theta - \sin \theta + \frac{\theta^2}{2} + \frac{\sin 2\theta}{4} - \cos \theta \right]_0^{\pi/2} = \pi/4$$

$$\int_{C3} dr = \int_0^1 (\pi/2, 0, 1) \cdot (0, 1, 0) dr = 0$$

$$\int_{C4} dr = \int_0^{\pi/2} (0, 0, 1) \cdot (0, 0, 1) dr = 0$$

$$\int_C F \cdot dr = \iint_S (\nabla \times \vec{F}) \cdot \vec{N} dS \rightarrow \text{Stokes theorem}$$

14. Sea $\mathbf{F}(x, y) = (xy, y^2)$ y sea c la trayectoria $y = 2x^2$ que une $(0,0)$ y $(1, 2)$ en

R2. Evaluar $\int_c \mathbf{F} \cdot d\mathbf{s}$. Depende la integral de la trayectoria que une el punto inicial y final?

$$r(t) = (t, 2t^2) \quad r'(t) = (1, 4t) \quad t \in [0, 1]$$

$$\begin{aligned} \int_c \mathbf{F}(x, y) \cdot d\mathbf{r} &= \int_0^1 \mathbf{F}(x(t), y(t)) \cdot r'(t) dt = \int_0^1 (2t^3, 4t^4) \cdot (1, 4t) dt \\ &= \int_0^1 2t^3 + 16t^5 dt = \left[\frac{1}{2}t^4 + \frac{16}{6}t^6 \right]_0^1 = \frac{19}{6} \end{aligned}$$

15. Sea $\mathbf{F}(x, y, z) = (2xyz + \sin x)\mathbf{i} + x^2z\mathbf{j} + x^2y\mathbf{k}$. Hallar una función f tal que $\mathbf{F} = \nabla f$.

El campo es conservativo si $\vec{\nabla} \times \vec{F} = 0$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz + \sin x & x^2z & x^2y \end{vmatrix} = x^2\vec{i} + 2xy\vec{j} + 2xz\vec{k} - 2xz\vec{k} - x^2\vec{i} - 2xy\vec{j} = 0$$

$$\vec{F} = \vec{\nabla} f$$

$$\int x = \int 2xy + \sin x \, dx = x^2yz - \cos x + g(y, z)$$

$$\int y = \int x^2z \, dy = x^2yz + g(x, z)$$

$$\int z = \int x^2y \, dz = x^2yz + g(x, y)$$

$$f(x, y, z) = x^2yz - \cos x$$

16. Mostrar que el siguiente campo vectorial es conservativo y calcular $\int_C \mathbf{F} \cdot d\mathbf{s}$ para la curva dada:

$$\mathbf{F} = \frac{2x}{y^2 + 1} \mathbf{i} - \frac{2y(x^2 + 1)}{(y^2 + 1)^2} \mathbf{j}$$

parametrización de la curva $x = t^3 - 1, y = t^6 - t, 0 \leq t \leq 1$

F es conservativo si $\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$

$$\left. \begin{aligned} \frac{\partial N}{\partial x} &= -\frac{4xy}{(y^2+1)^2} \\ \frac{\partial M}{\partial y} &= 2x \cdot -1 \cdot 2y \cdot (y^2+1)^{-2} = -\frac{4xy}{(y^2+1)^2} \end{aligned} \right\} \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y} \rightarrow F \text{ es conservativo}$$

$$F = \vec{\nabla} f$$

$$f_x = \int \frac{2x}{y^2+1} dx = \frac{x^2}{y^2+1} + g(y)$$

$$f_y = \int -\frac{2y(x^2+1)}{(y^2+1)^2} dy = \frac{x^2}{y^2+1} + \frac{1}{y^2+1} + g(x)$$

$$f(x,y) = \frac{x^2}{y^2+1} + \frac{1}{y^2+1} \rightarrow \int_a^b \vec{F} \cdot d\mathbf{r} = f(x(b), y(b)) - f(x(a), y(a))$$

$$\begin{aligned} x &\in [-1, 0] \\ y &\in [0, 0] \end{aligned} \quad (x,y) = [(-1, 0), (0, 0)]$$

$$\int F \cdot d\mathbf{r} = \frac{(0)^2+1}{0^2+1} - \frac{(-1)^2+1}{0^2+1} = 1 - 2 = -1$$

17. Calcular la integral de superficie $\iint_{\partial W} \mathbf{F} \cdot \mathbf{n} dA$ donde

$$\mathbf{F}(x, y, z) = \mathbf{i} + \mathbf{j} + z(x^2 + y^2)^2 \mathbf{k} \quad \partial W \text{ es la superficie del cilindro}$$
$$x^2 + y^2 \leq 1, 0 \leq z \leq 1$$

$$\iiint_V (\mathbf{F}(x, y, z) \cdot \vec{\nabla}) dV = \iiint_Q \vec{\nabla} \cdot \vec{F} dV$$

$$\vec{\nabla} \cdot \vec{F} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (1, 1, z(x^2 + y^2)^2) = 0 + 0 + (x^2 + y^2)^2$$

$$\iiint x^2 + y^2 dx dy dz \rightarrow \text{Paso a coordenadas cilíndricas}$$

$$x = r \cos \theta \quad r \in [0, 1]$$

$$y = r \sin \theta \quad \theta \in [0, 2\pi]$$

$$z = z \quad z \in [0, 1]$$

$$\int_0^1 \int_0^{2\pi} \int_0^1 r^4 \cdot r dr d\theta dz = \int_0^1 \int_0^{2\pi} \int_0^1 r^5 dr d\theta dz = \int_0^1 dz \cdot \int_0^{2\pi} d\theta \cdot \int_0^1 r^5 dr$$

$$= [z]_0^1 \cdot [\theta]_0^{2\pi} \cdot [r^6/6]_0^1 = 1 \cdot 2\pi \cdot 1/6 = \pi/3$$